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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S5 (R, S) / S5 (PT) (R,S) / S5 (WP) (R) Examination November 2024 (2019 Scheme)

Course Code: EET 305**Course Name: SIGNALS AND SYSTEMS**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- 1 Define unit step signal. Sketch the signal $x(t) = -2u(t + 2)$ (3)
- 2 Explain any two peculiar characteristics of non-linear systems. (3)
- 3 What are the conditions under which a periodic signal can be represented by a Fourier series? (3)
- 4 Find the Fourier transform of the rectangular pulse shown below in Fig.1 (3)

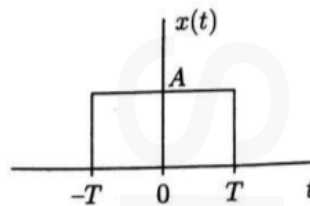


Fig.1

- 5 Determine the unit impulse response for the system with transfer function $H(s) = \frac{10}{s+100}$ (3)
- 6 List down the conditions under which a polynomial function is said to be positive real. (3)
- 7 Determine the transfer function of the Zero Order Hold circuit used for signal reconstruction. (3)
- 8 List down any 3 properties of discrete convolution. (3)
- 9 State and prove the time reversal property of discrete time Fourier series. (3)
- 10 Explain how bilinear transformation can be used to determine the stability of discrete-time systems. (3)

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- 11 a) Differentiate between energy and power signals. Determine whether the following signal, $x(t) = e^{-4t}u(t)$ is an energy signal or power signal. (7)

- b) Evaluate the convolution $y(t) = x_1(t) * x_2(t)$ where $x_1(t) = \sin 3t u(t)$ and $x_2(t) = u(t)$ (7)
- 12 a) Determine the fundamental period of the signal (5)
 $x(t) = 2\cos(10t + 1) - \sin(4t - 1)$
- b) With suitable examples, differentiate between (9)
 (a) Causal and non-causal systems
 (b) Static and dynamic systems
 (c) Stable and unstable systems

Module -2

- 13 a) Find the trigonometric Fourier Series for the signal shown below in Fig.2 (9)

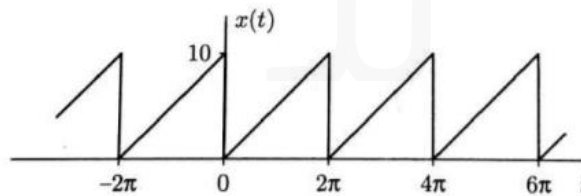


Fig.2

- b) State and prove the differentiation property of Fourier Transform. (5)
- 14 a) Derive the transfer function $\frac{X_2(s)}{F(s)}$ of the mechanical system shown below in Fig.3. (9)

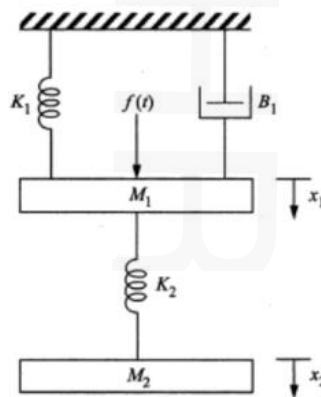


Fig.3

- b) Determine the zero-input response of a system represented by the differential equation $\frac{d^2y(t)}{dt^2} + 6\frac{dy(t)}{dt} + 6y(t) = 0$ using Laplace transform analysis. Assume that the initial conditions on the system are $y(0) = 1, \frac{dy(0)}{dt} = 2$ (5)

Module -3

- 15 a) Determine the overall transfer function using block diagram reduction. (5)

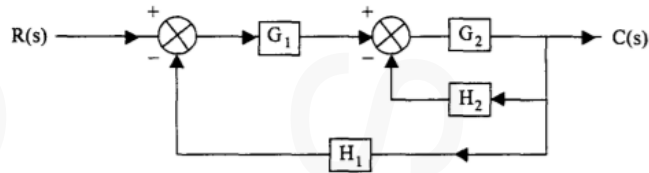


Fig.4

- b) Find the range of values of K for the closed loop system in below in Fig.5 to remain stable. (9)

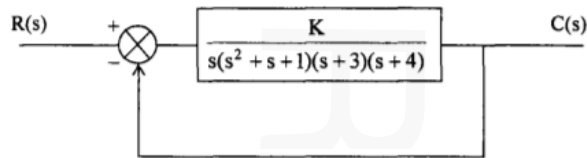


Fig.5

- 16 a) Determine the transfer function of the system represented by the SFG using Mason's Gain Formula. (9)

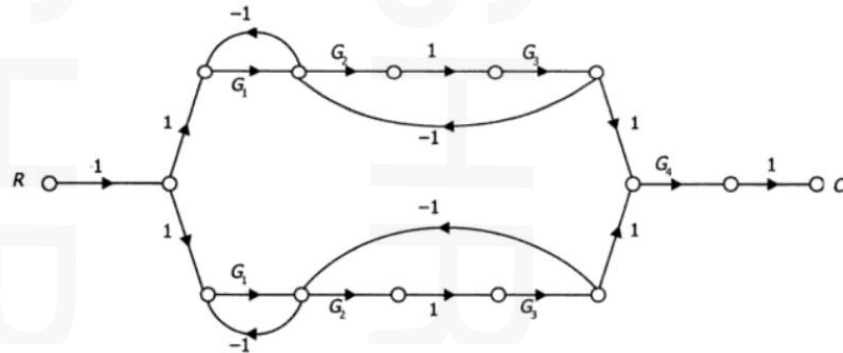


Fig.6

- b) Check the stability of the system represented by the following characteristic equation (5)
using Routh stability criterion: $s^4 + 2s^3 + 8s^2 + 4s + 3$

Module -4

- 17 a) Determine the convolution sum of the two sequences (8)
 $x(n) = \{1, 2, 3, 1\}$ and $h(n) = \{1, 2, 1, -1\}$
- b) Determine the Z-transform of $x(n) = e^{j\omega_0 n} u(n)$. Also, specify the ROC. (6)

- 18 a) Explain the aliasing effect in sampled data systems (5)
 b) Find the inverse Z-transform of (9)

$$X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$$

for the following three possibilities of ROC given as:

- (i) $|z| > 1$
 (ii) $|z| < 0.5$
 (iii) $0.5 < |z| < 1$

Module -5

- 19 a) A causal discrete time LTI system is described by (9)

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n)$$

where $x(n)$ and $y(n)$ are the input and output of the system, respectively.

Determine the transfer function $H(z)$ of the system. Also find the impulse response of the system.

- b) Determine the Fourier Transform of the sequence $x(n) = a^n u(n)$, $|a| < 1$ (5)
 20 a) Obtain the Direct form-II realization for the system described by the difference equation (9)

$$y(n) + \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) - \frac{7}{4}x(n-1) - \frac{1}{2}x(n-2)$$

- b) Check stability of the system described by the following characteristic equation, using (5)
 Jury's test: $2z^4 + 7z^3 + 10z^2 + 4z + 1$
